

Project Report
Modern Application Development – 2

Quiz Master – V2

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Project Details: -

It is an online quiz app for exam preparation for multiple courses. This is a two-user application, one is Administrator and other is regular user or student.

A site created for student to practice for their exam. They can practice subject wise to chapter wise. They can pick a chapter of their choice and start a quiz from all available quizzes that belongs to that chapter. They can attempt a particular quiz multiple times but there is a time limit, within one hours of completing a quiz they can not reattempt. They must wait for it to reattempt. Every quiz attempt is separately stored to give them an idea of how they are improving on a particular topic.

On other side Administrator or quiz-master who can create, update, read and delete quizzes and associated subjects and chapters. So, it is a two-role based application where admin can control quizzes and questions in it.

My Approach: -

First, I have studied the project statement document several times and tried to understand the problem and the core component of this project. I created a small rough document how this app is going to be. As a result, I created all the storage entity and relationship namely

- User – to store user details for authentication purpose
- Role – to authorization
- Admin – to store only admin information
- Student – to store only student information
- Subject – for subject of quiz
- Chapter – to store chapter under a subject for chapter wise evaluation
- Quiz – main quiz object that student will attempt
- Question – for storing questions from quizzes
- User-role – for connecting user with a role
- Scores – for storing every quiz attempt of student

and design and implemented the database schema of those entity and their relationship. Next, I worked on the authentication API system. I made the API and then create frontend UI for login and registration page.

After proper auth system I focus on Administrator page. I design that dashboard as main Vue component for admin and all child components of it. For example, a child component for creating subject, chapter and quiz and question, another child component for managing quizzes and questions, another for search feature and last one for summary of everything happening in the app. As I was building frontend UI I was building the APIs parallelly.

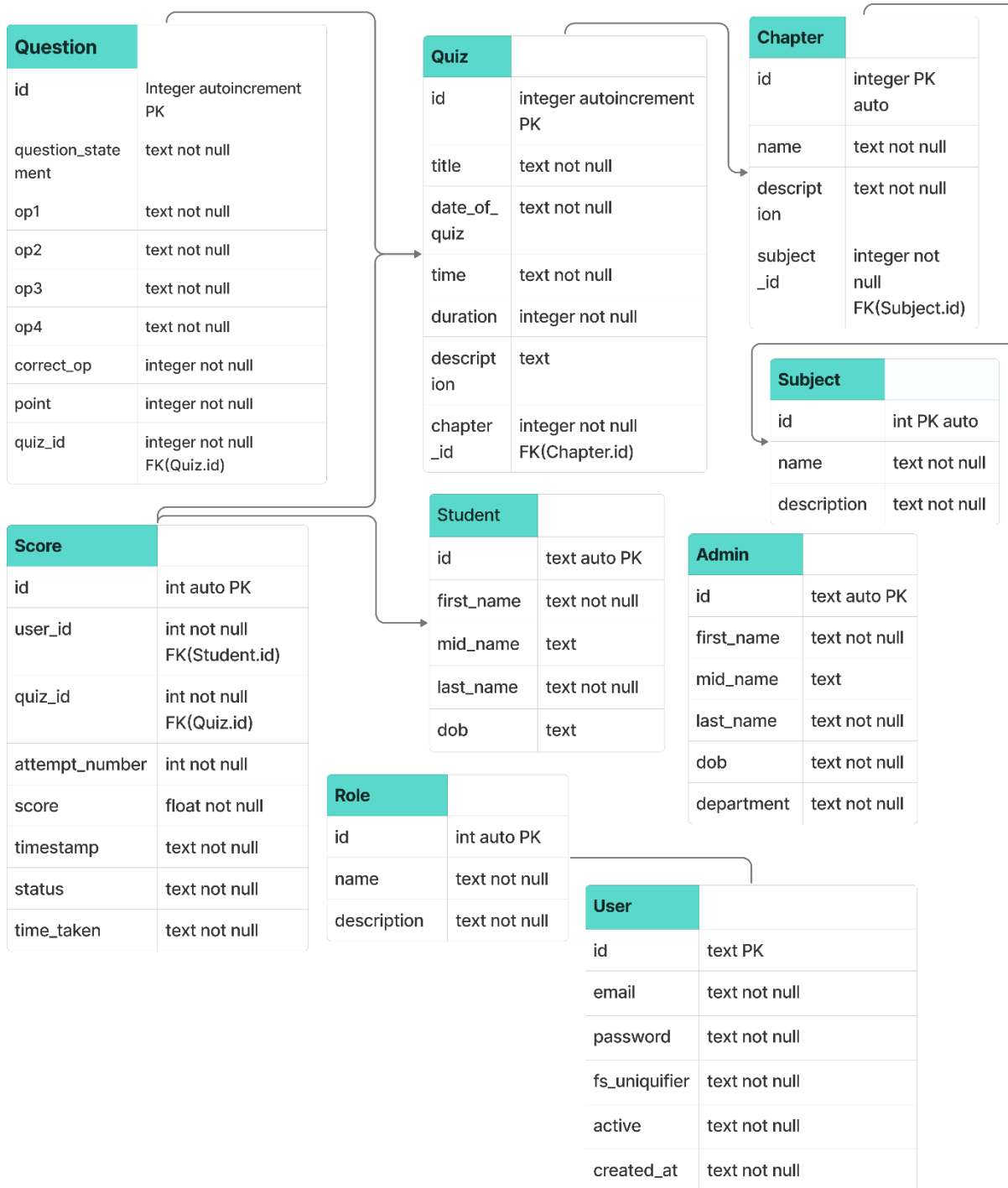
After completing Admin section, I shift my focus on student dashboard. It is also same, a main component for the dashboard and all child components. Three child components are here for home, profile, and summary of quiz attempt.

Frameworks and libraries used: -

The main frameworks and libraries that I have used is

- **Flask** - A python language-based web app development framework that I have used for application backend code.
- **Flask-Security-Too** – a flask extension for authentication and authorization system building. I have used token based with role security system.
- **Flask-SQLAlchemy** – also flask extension for working with ORM based database query system. I define all the database table using a object SQLAlchemy from this library.
- **SQLAlchemy** – Main python sqlalchemy library. I have used for attribute type highlighting in the model definition.
- **Pip** – for installing those packages.
- **Random** – for creating random id of newly registered student.
- **SQLite** – a database for storing all data that application generate.
- **VueJs** – a fronted reactive framework for User interface building.
- **Vue Router** – also a frontend routing framework to implement route mechanism at UI level.
- **Vue Store** – for storing global client data at UI
- **Bootstrap** – for styling the html page in vue template.

Entity-Relationship Diagram: -



Video Link-

https://drive.google.com/file/d/183efG2LOhiRMvPwBEOg8PpHTgYlitFcP/view?usp=drive_link